Teka-teki Angka (Number Puzzles & Missing Digits) - Yuk

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Student

Student: Yuki

Topic: Teka-teki Angka (Number Puzzles & Missing Digits)

Status: completed

Lesson Plan

Learning Objectives

- Identify the missing digit in a 3-digit addition equation using place value logic.
- Solve for missing digits in 3-digit subtraction equations without regrouping.
- Calculate the numerical value of symbols or letters in basic cryptarithm puzzles.
- Explain the deductive reasoning used to find a hidden number.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Yuki! Welcome to our Secret Agent Academy. To unlock your badge today, we need to pass the 'Security Clearance' check. First, let's look at our number scanner. Can you tell me what number is made of 4 hundreds, 2 tens, and 7 ones? Great! Now, look at this number: 835. How many 'tens' are hiding in there? Excellent. Now, let's look at some smaller numbers. If I have 12 gadgets and I lose 4, how many are left? And finally, what is 15 plus 5? You passed! Clearance granted!

Activities:

- *Place Value Scanner: Identify 427 from expanded form.*
- *Digit Finder: Identify the value of the '3' in 835.*
- *Quick-fire 1-20: Solve 12 - 4.*
- *Quick-fire 1-20: Solve 15 + 5.*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Today, Yuki, you are a Code Breaker! Imagine a spy sent us a secret message, but some of the ink spilled on the numbers. We need to use our math powers to see 'through' the ink. Being able to find missing numbers helps us solve mysteries and fix mistakes in real life. Are you ready to find the invisible digits?

Activities:

- *Storytelling: The 'Ink Spill' Mystery.*
- *Visualizing missing parts in a whole.*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Before we solve the 3-digit puzzles, let's remember our secret weapon: Place Value. In the number 145, the 5 is a 'One', the 4 is a 'Ten', and the 1 is a 'Hundred'. When we add or subtract, we always check the 'Ones' column first. It's like checking the lock on the door before entering the house!

Activities:

- *Column Alignment: Drawing lines between H-T-O columns on the whiteboard.*

4. Problem Introduction (10 minutes)

Teacher Script:

Look at this puzzle: 12[?] + 1[?]4 = 259. Oh no! The ink covered two digits! Let's look at the 'Ones' column. [?] + 4 = 9. What number plus 4 equals 9? Use your fingers or mental math. Yes, 5! Now let's look at the 'Tens'. 2 + [?] = 5. What's the missing link? Yes, 3! We just cracked the code!

Activities:

- *Step-by-step breakdown of a vertical addition puzzle.*
- *Introduction to the 'Box' symbol as a placeholder.*

5. Guided Practice (15 minutes)

Teacher Script:

Now it's your turn to lead the mission. I'll be your assistant. Let's try a subtraction puzzle: 587 - [?][?]2 = 235. Start with the Ones. 7 minus what is 5? Good, 2. Now the Tens: 8 minus what is 3? Yes, 5. And the Hundreds: 5 minus what is 2? You got it, 3! The missing number was 352. You're a natural!

Activities:

- *Interactive dragging of digit cards into blank boxes.*
- *Subtraction mystery: 456 - 1[?]2 = 324.*

6. Independent Practice (15 minutes)

Teacher Script:

Agent Yuki, here is a Top Secret folder with 3 puzzles. One is a 'Symbol Code' where a Star represents a number. Can you solve them all by yourself in the chatbox? I'm watching the perimeter!

Activities:

- *Puzzle 1: 24[?] + 3[?]1 = 579.*
- *Puzzle 2: 9[?]5 - [?]21 = 614.*
- *Puzzle 3 (Cryptarithm): A + A = 6. What is A? Then solve: 12A + 11A = ?*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Mission Accomplished! You found all the missing digits. What was your favorite part—adding or subtracting? Remember, when a number is missing, just look at its neighbors in the column. Before you go, don't forget to tell your parents to help you log in to https://learn.algonova.id/login to see your special agent homework. See you next time!

Activities:

- *Self-assessment (Thumbs up/down).*
- *Verbal summary of the 'column rule'.*

Homework

Medium Questions:

1. Find the missing digit: 34[?] + 122 = 469
2. Find the missing digit: 5[?]7 - 213 = 354
3. Find the missing digit: [?]25 + 234 = 959
4. If A + 3 = 10, what is A? Use this to solve: 14A + 101 = ?

Hard Questions:

1. Solve for the two missing digits: 1[?]2 + [?]35 = 477
2. Solve for the two missing digits: 8[?]9 - 42[?] = 432
3. Cryptarithm: If Heart + Heart = 8, what is 2Heart4 + 111?
4. Double Mystery: [?][?]9 - 111 = 558
5. Identify the missing digit in the tens and hundreds: [?]4[?] + 2[?]1 = 599
6. Advanced Logic: If A = 2 and B = 5, find the missing digit in 1AB + 2[?]1 = 396

Answer Key:

`34[7] + 122 = 469 (7+2=9)`

`5[6]7 - 213 = 354 (6-1=5)`
`[7]25 + 234 = 959 (7+2=9)`
`A=7; 147 + 101 = 248`
`1[4]2 + [3]35 = 477`
`8[5]9 - 42[7] = 432`
`Heart=4; 244 + 111 = 355`
`[6][6]9 - 111 = 558`
`[3]4[8] + 2[5]1 = 599`
`125 + 2[7]1 = 396 (2+7=9)`
REVIEW: Place Value - `300 + 40 + 2 = 342`
REVIEW: Place Value - `Digit 7 in 712 is 700`
REVIEW: Place Value - `Largest number from 3, 1, 8 is 831`
REVIEW: Place Value - `5 tens and 0 ones is 50`
REVIEW: Numbers 1-20 - `17 - 9 = 8`
REVIEW: Numbers 1-20 - `14 + 6 = 20`
REVIEW: Numbers 1-20 - `Two more than 18 is 20`
REVIEW: Numbers 1-20 - `Half of 12 is 6`

Parent Reminder

Ø=Ýuþ &Bþ Mission Accomplished: Agent Yuki Cracked the Code!

Yuki did a fantastic job today as a Math Detective! We explored 3-digit addition and subtraction by finding 'hidden' numbers. She is starting to use logic to solve math puzzles. Please encourage her to complete the 'Secret Mission' homework on the portal: https://learn.algonova.id/login